

Financial Modeling and Dealing FN585

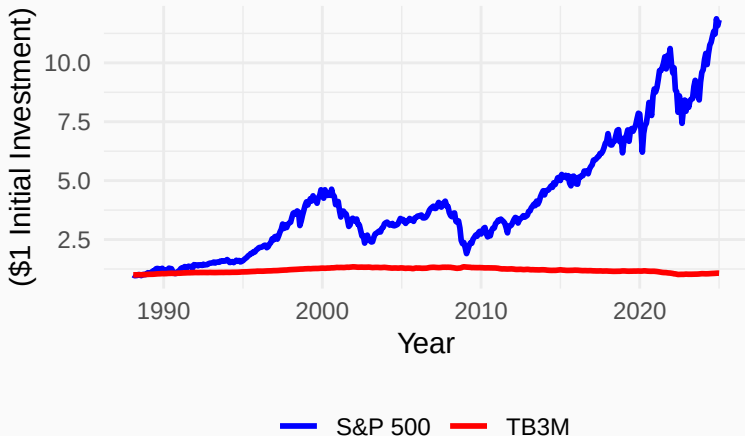
Financial Data and Their Characteristics

Shahabeddin Gharaati

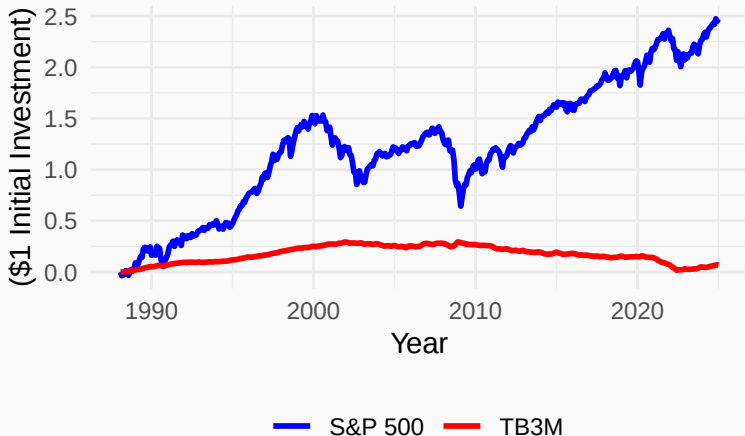
University of Brighton

Stilized Fact

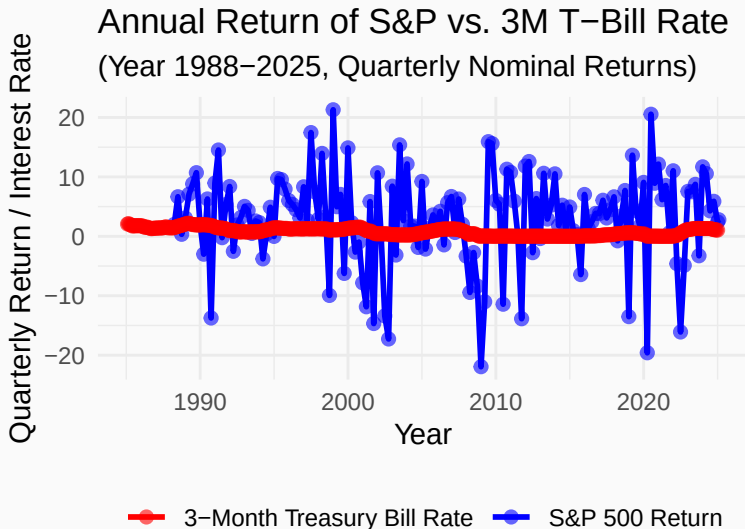
S&P 500 vs. 3-Month T-Bills (Inflation-Adjusted)



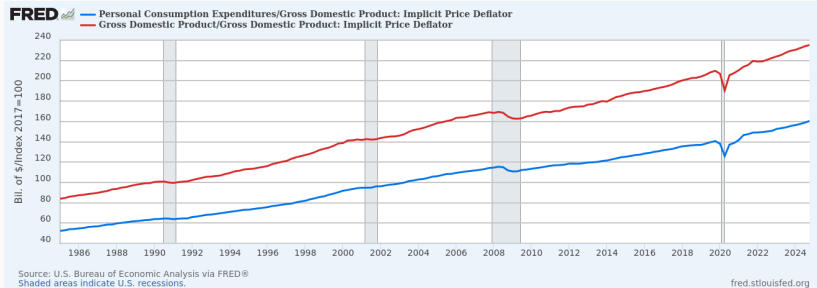
S&P 500 vs. 3-Month T-Bills (Inflation-Adjusted)



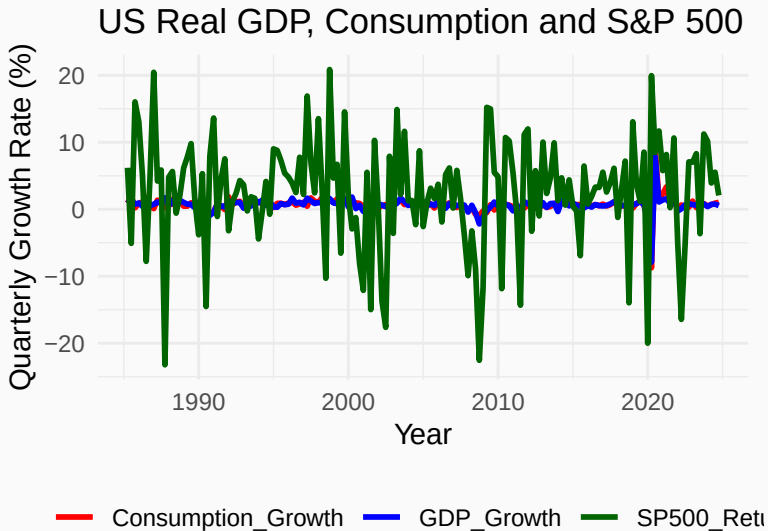
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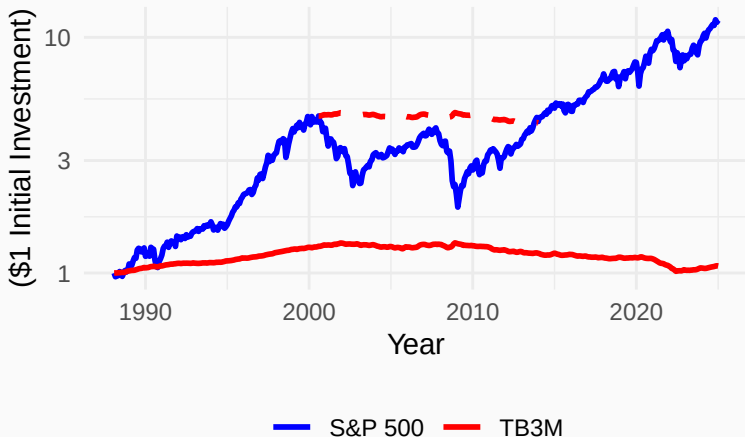
More Stylized Fact from 1985-2025

Table 1: Table: Summary Statistics. μ = expectation, σ = standard deviation, and ρ = correlation with stock. Significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

	s&p 500	T-Bill	Risk Premium	GDP	Consumption
μ :	10.72	3.19	7.53	5.10	5.35
σ :	16.6	2.5	16.4	2.33	2.56
ρ :	1	0.15	0.98***	0.07	0.00

More Stilized Fact

\$1 Invested in S&P 500 vs. 3-Month T-Bills
(Inflation-Adjusted)



What are Asset Returns?

- **Definition:** Return on an asset measures the change in its value over a period, expressed as a percentage of the initial value.
- **Significance:** Fundamental metric in finance, reflecting the investment's profitability or loss.

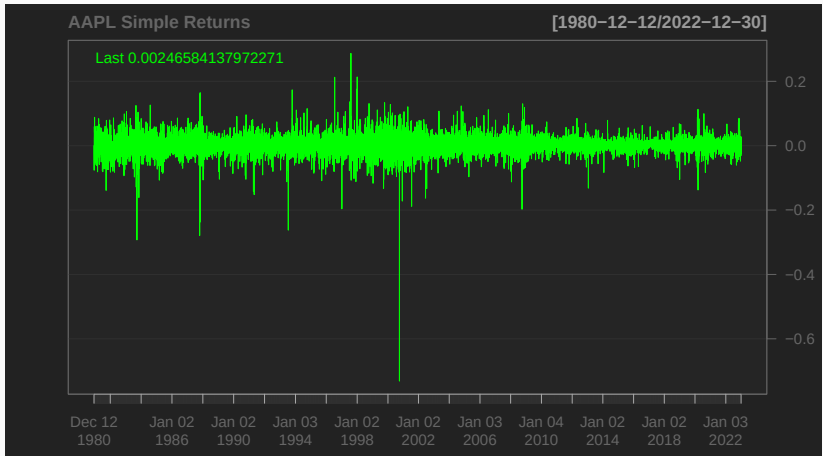
Simple vs. Log Returns

- **Simple Returns:** Suitable for single-period investment analysis.
 - Formula: $R_t = \frac{P_t - P_{t-1}}{P_{t-1}}$
- **Logarithmic Returns:** Preferred for multiple-period analysis due to additive property.
 - Formula: $r_t = \log\left(\frac{P_t}{P_{t-1}}\right) = \log(P_t) - \log(P_{t-1})$

Calculating Simple Returns in R

Example: Apple Inc. Stock

```
## [1] "AAPL"
```



One-period simple net return:

The one-period simple net return (R_t) is calculated as:

$$R_t = \frac{P_t}{P_{t-1}} - 1$$

where P_t is the price of an asset at time t .

One-period continuously compounded (or log) return:

The one-period continuously compounded return (r_t) is defined as:

$$r_t = \ln(1 + R_t) = \ln\left(\frac{P_t}{P_{t-1}}\right) = p_t - p_{t-1}$$

where $p_t = \ln(P_t)$. This is based on the concept of continuous compounding:

$$\lim_{m \rightarrow \infty} \left(1 + \frac{r}{m}\right)^m = e^r = 1 + R \implies r = \ln(1 + R)$$

Advantages of returns over prices:

- Normalization
- Returns have more attractive statistical properties (stationarity)

k-period simple gross return:

The k-period simple gross return ($\widetilde{R}_t(k)$) is calculated as follows:

$$\begin{aligned}1 + R_t(k) &= \frac{P_t}{P_{t-k}} = \frac{P_t}{P_{t-1}} \cdot \frac{P_{t-1}}{P_{t-2}} \cdots \frac{P_{t-k+1}}{P_{t-k}} \\&= (1 + R_t)(1 + R_{t+1}) \cdots (1 + R_{t-k+1}) \\ \widetilde{R}_t(k) &= \left(\prod_{i=0}^{k-1} (1 + R_{t-i}) \right)^{\frac{1}{k}} - 1\end{aligned}$$

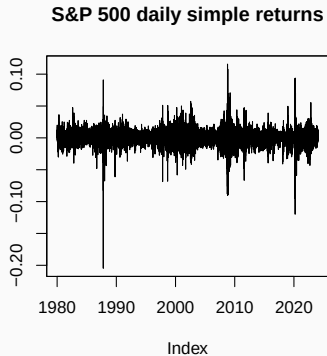
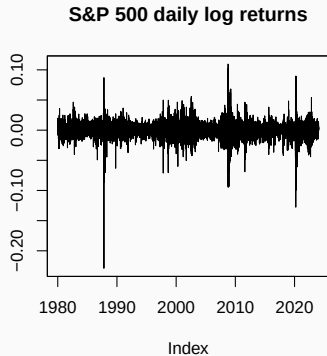
k-period log-return:

The k-period log-return ($r_t(k)$) is defined as:

$$\begin{aligned} r_t(k) &= \ln(1 + R_t(k)) = \ln((1 + R_t)(1 + R_{t+1}) \cdots (1 + R_{t-k+1})) \\ &= \ln(1 + R_t) + \ln(1 + R_{t+1}) + \cdots + \ln(1 + R_{t-k+1}) \\ &= r_t + r_{t-1} + \cdots + r_{t-k+1} \Rightarrow \bar{r}_t(k) = \frac{1}{k} \sum_{j=0}^{k-1} r_{t-j} \end{aligned}$$

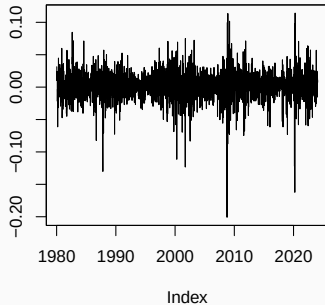
The reported time interval over which are returns calculated is usually one day, one month, one quarter, one year.

Time series of S&P500 simple and log daily returns

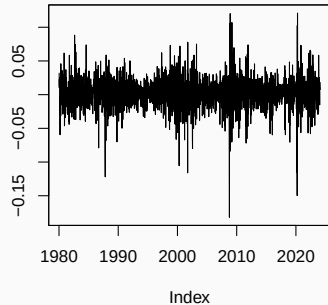


Time series of S&P500 simple and log weekly returns

S&P 500 weekly log returns

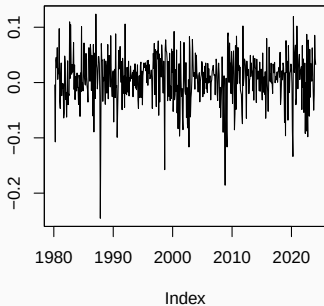


S&P 500 weekly simple returns

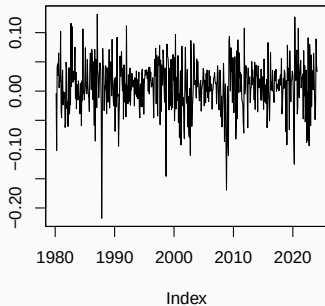


Time series of S&P500 simple and log monthly returns

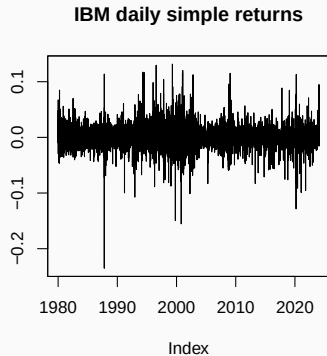
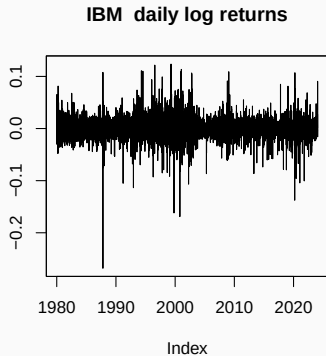
S&P 500 monthly log returns



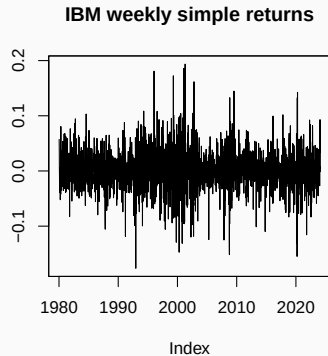
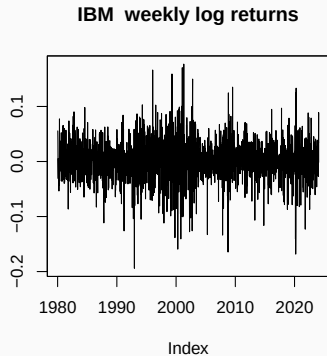
S&P 500 monthly returns



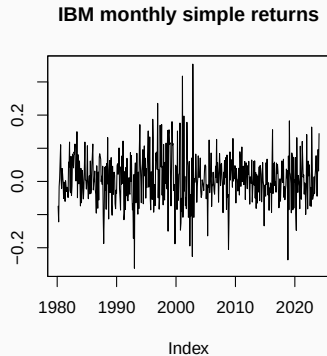
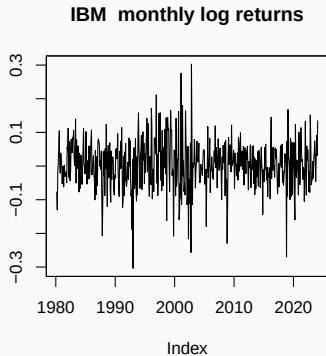
Time series of IBM simple and log daily returns



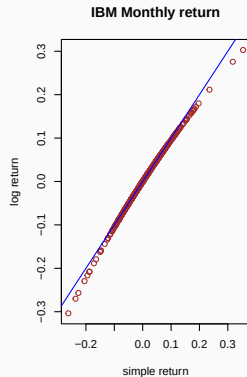
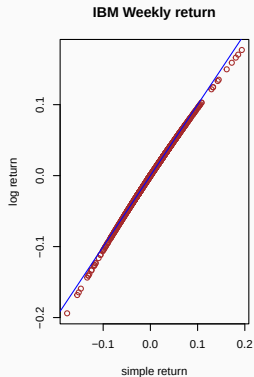
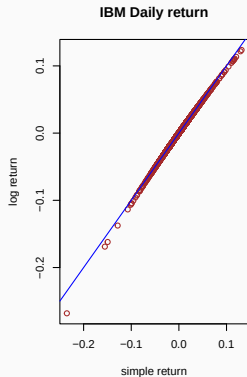
Time series of IBM simple and log weekly returns



Time series of IBM simple and log monthly returns



Time series of IBM simple and log returns



Notations

Definition (distribution of returns):

We represent the (log-) returns by the real-valued random variable R_t (or $R_t[k]$). The distribution of R_t is called the **distribution of returns**, or returns distribution.

- The **probability distribution function (pdf)** of R_t is denoted $P_R(R_t = r)$.

$$P(R_t \leq r) = \sum_{s \in \{s: X_s \leq r\}} P_R(x_s)$$

Notations

Definition (distribution of returns):

We represent the (log-) returns by the real-valued random variable R_t (or $R_t[k]$). The distribution of R_t is called the **distribution of returns**, or returns distribution.

- The **probability density function (pdf)** of R_t is denoted $f_R(r)$.
- The **cumulative density function (cdf)** of R_t is denoted $F_R(r)$ such that:

$$P(R_t \leq r) = F_R(r) = \int_{-\infty}^r f_R(x) dx$$

Statistical properties of financial data

Example (Normality assumption)

Let us assume that the return R_t has a normal distribution

$$R_t \sim \mathcal{N}(\mu, \sigma^2)$$

with μ the expected (mean of) return and σ^2 the variance of return.

Then, the pdf and cdf of R_t are respectively defined by

$$f_R(r) = \frac{1}{\sigma} \phi\left(\frac{r - \mu}{\sigma}\right) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(r - \mu)^2}{2\sigma^2}\right)$$

$$F_R(r) = \Phi\left(\frac{r - \mu}{\sigma}\right) = \int_{-\infty}^r f_R(u) du$$

where $\phi(\cdot)$ and $\Phi(\cdot)$ respectively denote the pdf and cdf of the standard normal distribution.

Statistical properties of financial data

The distribution of returns can be characterized either by its pdf or by its **moments**.

Definition (non-central and central moments)

The k th **non-central moment** of R_t is given by

$$m_k = \mathbb{E} [R_t^k] = \int_{-\infty}^{\infty} x^k f_R(x) dx = \sum_{s=1}^S x_s^k P_R(x_s = x)$$

The k th **central moment** of R_t is given by

$$\mu_k = \mathbb{E} [(R_t - m_1)^k] = \int_{-\infty}^{\infty} (x - m_1)^k f_R(x) dx = \sum_{s=1}^S (x_s - m_1)^k P_R(x_s)$$

with $m_1 = \mathbb{E}(R_t)$, the mean of the return distribution.

Statistical properties of financial data

In financial econometrics, we are mainly interested in the first four moments:

Mean Variance Skewness Kurtosis

Table 2

Descriptive statistics of deseasonalized data for Royal Bank (RY), Placer Dome (PDG), and Alcan (AL) stocks.

	Mean	Std. dev	Skew	Kurt	Max	Min	$Q(15)$	$Q_2(15)$
Durations								
RY	0.99	1.14	2.55	12.73	15.97	0.01	1585	480
PDG	1.03	1.59	9.22	289.14	81.47	0.01	2547	68
AL	1.00	1.48	3.59	22.81	21.39	0.01	5628	1477
Returns								
RY	0.001	0.983	-0.001	7.310	6.889	-8.391	8155	14,690
PDG	0.002	0.991	0.025	6.752	7.860	-6.907	2425	3112
AL	0.007	0.984	-0.200	11.521	9.195	-19.795	4323	3432

Mean is the sample mean, *Std. dev* is the sample standard deviation, *Skew* is the sample skewness, *Kurt* is the sample kurtosis, *Max* is the sample maximum, *Min* is the sample minimum. $Q(15)$ is the Ljung-Box test statistics with 15 lags, $Q_2(15)$ is the Ljung-Box test statistics applied to squared variables using 15 lags. The associated 95% critical value is 24.99.

Source: Dionne G., Duchesne, P., and Pacurar, M. (2005), Intraday Value at Risk (IVaR) using tick-by-tick data with application to the Toronto Stock Exchange, *Journal of Empirical Finance*, 16(5), 777-792.

Mean and Variance of Returns

The **mean** (expected value) of the return R_t is represented as:

$$\mu = m_1 = \mathbb{E}(R_t) = \int_{-\infty}^{\infty} x f_R(x) dx$$

The **variance** of the return R_t is given by:

$$\sigma^2 = \mathbb{V}(R_t) = \mathbb{E}((R_t - \mu)^2) = \int_{-\infty}^{\infty} (x - \mu)^2 f_R(x) dx$$

Where $f_R(x)$ is the probability density function of returns.

Definition of Volatility

Volatility is a statistical measure of the dispersion of returns for a given security or market index. It can be measured by using the standard deviation of historical returns.

$$\text{Volatility} \equiv \sigma = \sqrt{V(R_t)}$$

Here, $V(R_t)$ represents the variance of the return R_t .

Statistical properties of financial data

Definition of Skewness Coefficient

Skewness is a measure of the asymmetry of the probability distribution of a real-valued random variable. It is the third central moment and is defined as follows:

$$\mu_3 = \mathbb{E} [(R_t - \mu)^3] = \int_{-\infty}^{\infty} (x - \mu)^3 f_R(x) dx = \sum_{s=1}^S (x_s - \mu)^3 P_R(x_s)$$

The standardized **skewness coefficient** is calculated by dividing the third central moment by the cube of the standard deviation (σ), which normalizes the measure:

$$S(R_t) = \mathbb{E} \left[\left(\frac{R_t - \mu}{\sigma} \right)^3 \right] = \frac{\mu_3}{\sigma^3}$$

Statistical properties of financial data

Where μ is the mean of the return distribution and $f_R(x)$ is its probability density function.

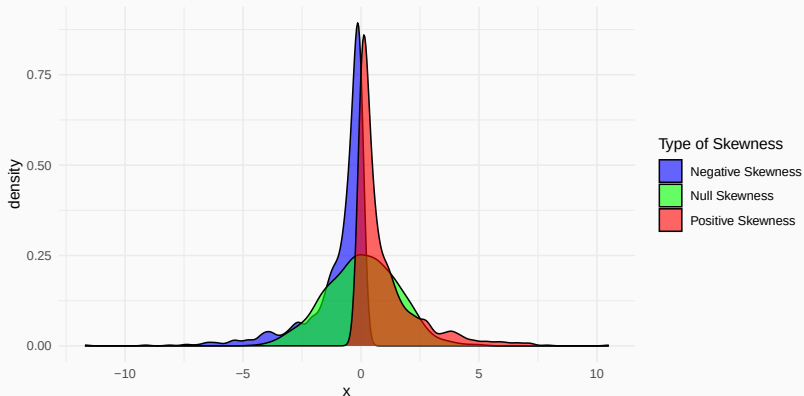
Skewness

- When $S(R_t) < 0$, large realizations of R_t are more often negative than positive (i.e., crashes are more likely than booms).
- The distribution is said to be **left-skewed**, **left-tailed**, or **skewed to the left**.
- For symmetric distributions (e.g., normal, Student, etc.), the skewness coefficient is null $S(R_t) = 0$.

Statistical properties of financial data

Figure: Illustration of the Skewness

The chart below shows three different distributions to illustrate positive skewness, null skewness, and negative skewness.



Definition of Kurtosis Coefficient

The fourth central moment is used to measure **kurtosis** and is defined as:

$$\mu_4 = \mathbb{E} [(R_t - \mu)^4] = \int_{-\infty}^{\infty} (x - \mu)^4 f_R(x) dx = \sum_{s=1}^S (x_s - \mu)^4 P_R(x_s)$$

The standardized **kurtosis coefficient**, which helps in comparing different distributions, is calculated by dividing the fourth central moment by the fourth power of the standard deviation (σ):

$$K(R_t) = \mathbb{E} \left[\left(\frac{R_t - \mu}{\sigma} \right)^4 \right] = \frac{\mu_4}{\sigma^4}$$

Statistical properties of financial data

This normalized measure of kurtosis is sometimes referred to as “excess kurtosis,” since it compares the kurtosis of the distribution to that of a normal distribution, which has a kurtosis of 3.

Kurtosis: properties

Fact (normal distribution)

For a normal distribution $R_t \sim \mathcal{N}(\mu, \sigma^2)$, the kurtosis coefficient is equal to 3

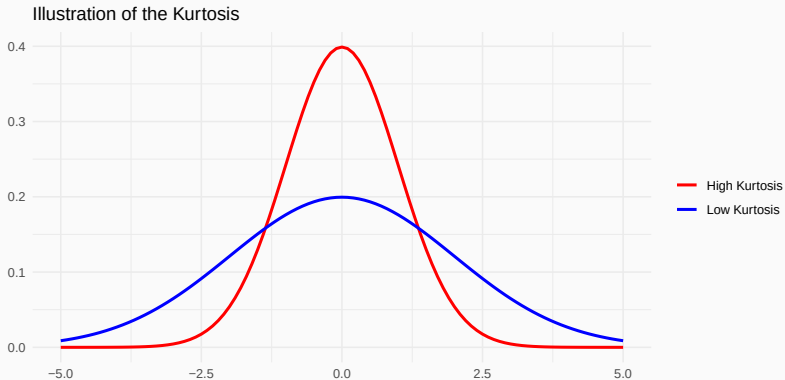
$$\text{Kurtosis} \quad K(R_t) = 3$$

Note: the Gaussian distribution is not the only one to have a kurtosis equal to 3.

Kurtosis: properties (cont'd)

- Large $K(R_t)$ implies that large realizations (positive or negative) are more likely to occur (compared to a normal distribution)
- If $K(R_t) > 3$, the distribution of R_t is said to be **leptokurtic**.
- If $K(R_t) = 3$, the distribution of R_t is said to be **mesokurtic**.
- If $K(R_t) < 3$, the distribution of R_t is said to be **platykurtic**.
- The excess kurtosis coefficient is equal to $K(R_t) - 3$.

Statistical properties of financial data



Statistical properties of financial data

Summary

Each (central or non-central) moment gives information on the pdf of returns

Moment	Formula	Interpretation
Mean	$\mathbb{E}(R_t) = \mu$	Indicator of central tendency
Variance	$\mathbb{V}(R_t) = \mathbb{E}((R_t - \mu)^2) = \sigma^2$	Indicator of dispersion around μ
Skewness	$S(R_t) = \mathbb{E}((R_t - \mu)^3)/\sigma^3$	Indicator of symmetry
Kurtosis	$K(R_t) = \mathbb{E}((R_t - \mu)^4)/\sigma^4$	Indicator of tail heaviness

Example (normal distribution)

If we assume that the returns are normally distributed with $R_t \sim \mathcal{N}(\mu, \sigma^2)$, then

$$\mathbb{E}(R_t) = \mu$$

$$\mathbb{V}(R_t) = \sigma^2$$

$$\mathbb{S}(R_t) = 0$$

$$\mathbb{K}(R_t) = 3$$

The normal distribution is **symmetric** and by definition, **mesokurtic**.

Statistical properties of financial data

Example (Student distribution)

If we assume that the return R_t has a Student distribution with ν degrees of freedom, i.e. $R_t \sim t(\nu)$, then we have

$$\mathbb{E}(R_t) = 0$$

$$\mathbb{V}(R_t) = \frac{\nu}{\nu - 2} \quad \text{if } \nu > 2$$

$$\mathbb{S}(R_t) = 0$$

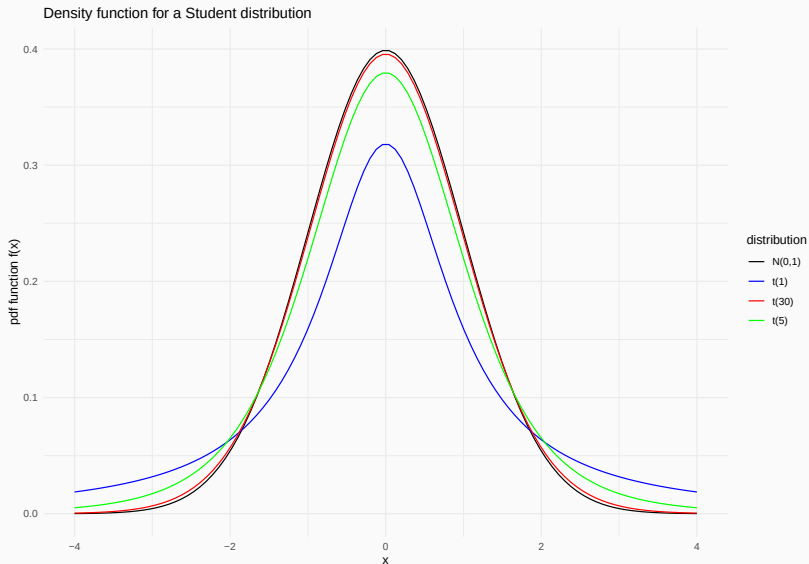
$$\mathbb{K}(R_t) = 3 - \frac{6}{\nu - 4} \quad \text{if } \nu > 4$$

The Student distribution is **symmetric** and **leptokurtic** (if ν is “small”). Its excess of Kurtosis decreases with the number of degrees of freedom. When ν tends to ∞ , $t(\nu) \approx \mathcal{N}(0, 1)$, and the distribution is mesokurtic.

Remarks

- The moments allow to characterize the shape of the returns distribution. However, the (theoretical) moments are unobservable and we need to estimate them.
- Denote by $\{R_1, \dots, R_T\}$ a sample of i.i.d. variables that have the same distribution as R_t .
- Denote by $\{r_1, \dots, r_T\}$ the realization of this sequence $\implies$ your dataset.

Statistical properties of financial data



Estimator Definition

An estimator is any function of a sample $H(R_1, R_2, \dots, R_T)$. Any descriptive statistic is an estimator.

Example: Sample Mean

- Assume i.i.d. random variables $R_1, R_2, \dots, R_T$.
- Sample mean: $\hat{\mu}_T = \frac{1}{T} \sum_{t=1}^T R_t$.
- Estimates the theoretical mean $\mu = E(R_t)$.

Sample Variance

- Assume i.i.d. random variables $R_1, R_2, \dots, R_T$.
- Sample variance: $\hat{\sigma}_T^2 = \frac{1}{T-1} \sum_{t=1}^T (R_t - \bar{R}_T)^2$.
- Estimator for the theoretical variance $\sigma^2 = V(R_t)$.
- Corrected for small sample bias.

Sample Skewness

- Assume i.i.d. random variables $R_1, R_2, \dots, R_T$.
- Sample skewness coefficient: $\hat{S}_T = \frac{1}{T} \sum_{t=1}^T \left(\frac{R_t - \bar{R}_T}{\hat{\sigma}} \right)^3$.
- Estimator for the theoretical skewness $S(R_t)$.

Sample Kurtosis

- Assume i.i.d. random variables $R_1, R_2, \dots, R_T$.
- Sample kurtosis coefficient: $\hat{K}_T = \frac{1}{T} \sum_{t=1}^T \left(\frac{R_t - \bar{R}_T}{\hat{\sigma}} \right)^4$.
- Estimator for the theoretical kurtosis $K(R_t)$.

Sampling Distribution

- An estimator $\hat{\theta}$ is a random variable.
- It has a probability distribution characterized by a pdf $f_{\hat{\theta}}(u)$.

Definition

- The distribution of an estimator or a statistic is called the sampling distribution.
- Characterized by moments such as expectation and variance.

Statistical properties of financial data

Definition (estimate)

A (point) estimate is the realized value of an estimator (i.e. a number) that is obtained when a sample is actually taken. For an estimator $\hat{\theta}$ it can be denoted by $\hat{\theta}(y)$.

Example (Point estimate)

For instance, a realization of the estimator R_T , denoted r_T , is defined as

$$r_T = \frac{1}{T} \sum_{t=1}^T r_t$$

- If $T = 3$ and $\{r_1, r_2, r_3\} = \{3, -1, 2\}$ then $r_T = 1.333$.
- If $T = 3$ and $\{r_1, r_2, r_3\} = \{4, -8, 1\}$ then $r_T = -1$.
- etc..

Statistical properties of financial data

- **Sample skewness:**

$$\hat{S}(x) = \frac{1}{(T-1)\hat{\sigma}^3} \sum_{t=1}^T (x_t - \hat{\mu}_x)^3,$$

- **Sample kurtosis:**

$$\hat{K}(x) = \frac{1}{(T-1)\hat{\sigma}^4} \sum_{t=1}^T (x_t - \hat{\mu}_x)^4.$$

Normality Assumption

Under normality assumption,

$$\hat{S}(x) \sim N\left(0, \frac{6}{T}\right), \quad \hat{K}(x) - 3 \sim N\left(0, \frac{24}{T}\right).$$

- **Test for symmetry:**

$$S^* = \frac{\hat{S}(x)}{\sqrt{6/T}} \sim N(0, 1)$$

if normality holds.

Decision Rule:

Reject H_0 of a symmetric distribution if $|S^*| > Z_{\alpha/2}$ or p-value is less than α .

Test for Tail Thickness

- Test for tail thickness:

$$K^* = \frac{\hat{K}(x) - 3}{\sqrt{24/T}} \sim N(0, 1)$$

if normality holds.

Decision Rule:

Reject H_0 of normal tails if $|K^*| > Z_{\alpha/2}$ or p-value is less than α .

Jarque-Bera Test

- A joint test (Jarque-Bera test):

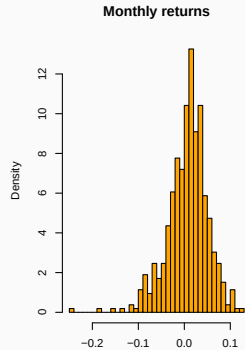
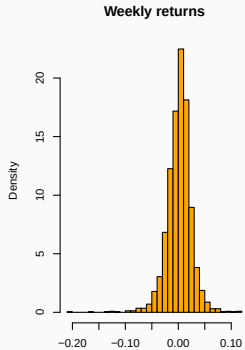
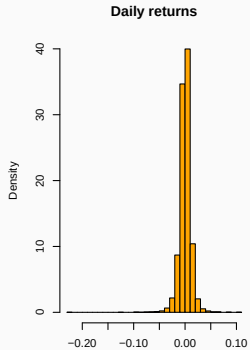
$$JB = (K^*)^2 + (S^*)^2 \sim \chi_2^2$$

if normality holds, where χ_2^2 denotes a chi-squared distribution with 2 degrees of freedom.

Decision Rule:

Reject H_0 of normality if $JB > \chi^2(\alpha)$ or p-value is less than α .

Histograms of S&P 500 Return



Assumptions on distributions of returns

- Normal distribution
- Lognormal distribution
- Scale mixture of normal distributions
- Other extended parametric, semi-parametric, non-parametric alternatives.

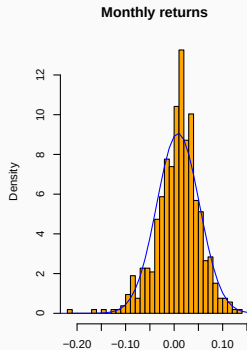
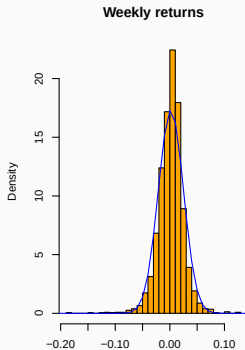
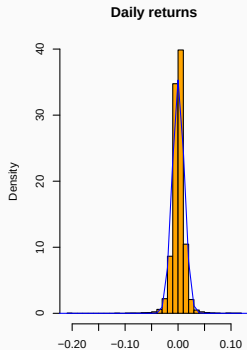
Normal distribution of simple returns

Assumption: Simple returns are independently and identically distributed as **normal** $\Rightarrow$ tractable statistical properties.

Disadvantages:

- No lower bound on simple asset returns (which is -1)
- k -period simple return (a product of one-period simple normal returns) is not normally distributed
- Normality assumption is not supported by the empirical evidence: many asset returns have fat tailed distribution (positive excess kurtosis)

Histograms of S&P 500 simple Return



QQ-plot of S&P 500 Return

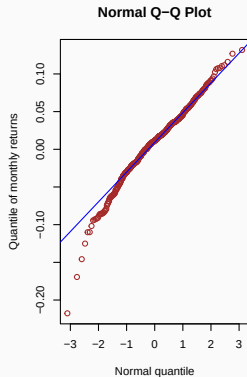
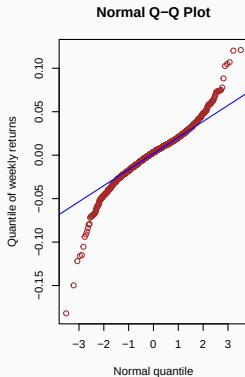
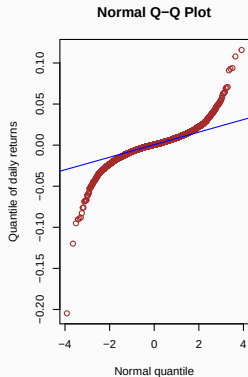


Table 3: Critical Value is 7.18

S&P 500	JB Statistic	p-Value
Daily returns	170748.0997	0
Weekly returns	2718.6954	0
Monthly returns	103.1002	0

Lognormal distribution of simple returns

Assumption: log returns are independently and identically distributed (iid) as normal $\Rightarrow$ simple returns are iid **lognormal**

- Advantage: No lower bound on log returns but lower bound on simple returns as $R_t = \exp(r_t) - 1 \geq -1$
- Advantage: k -period log return (a sum of one-period log normal returns) is normally distributed
- Disadvantage: Returns drawn from a fat-tailed distribution are not supported by this assumption

Stylized features of financial returns

Stationarity

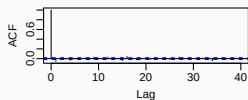
The prices of an asset recorded over times are often not stationary due to, for example, the steady expansion of economy, the increase of productivity resulting from technology innovation, and economic recessions or financial crisis. However their returns, denoted by r_t for $t \geq 1$, typically fluctuates around a constant level, suggesting a constant mean over time.

In fact most return sequences can be modeled as a stochastic processes with at least time-invariant first two moments (i.e. the weak stationarity)

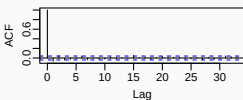
Stylized features of financial returns

ACF plots of the S&P500

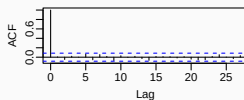
Daily returns



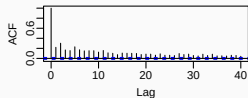
Weekly returns



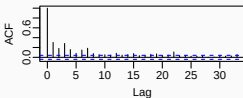
Monthly returns



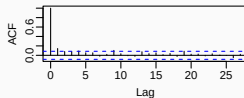
Squared daily returns



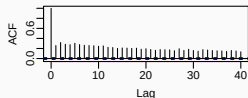
Squared weekly returns



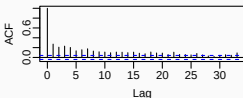
Squared monthly returns



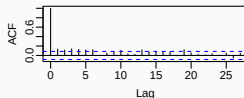
Absolute daily returns



Absolute weekly returns



Absolute monthly returns



Stylized features of financial returns

Heavy tails

The probability distribution of return r_t often exhibits heavier tails than those of a normal distribution.

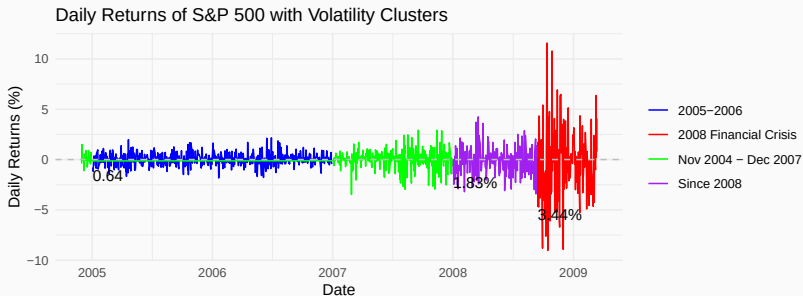
Asymmetry

The distribution of return r_t is often negatively skewed, reflecting the fact that the downturns of financial markets are often much steeper than the recoveries. Investors tend to react more strongly to negative news than to positive news.

Stylized features of financial returns

Volatility clustering.

This term refers to the fact that large price changes (i.e. returns with large absolute values) occur in clusters. Indeed, large price changes tend to be followed by large price changes, and periods of tranquility alternate with periods of high volatility.



Aggregational Gaussianity

Note that a return over k days is simply the aggregation of k daily returns. When the time horizon increases, the central limit theorem sets in and the distribution of the returns over a long time-horizon (such as a month) tends toward a normal distribution.

Stylized features of financial returns

Long range dependence.

- The returns themselves hardly show any serial correlation, which, however, does not mean that they are independent.
- daily squared and absolute returns often exhibit small and significant autocorrelations. Those autocorrelations are persistent for absolute returns, indicating possible long-memory properties.
- It is also noticeable that those autocorrelations become weaker and less persistent when the sampling interval is increased from a day, to a week to a month.

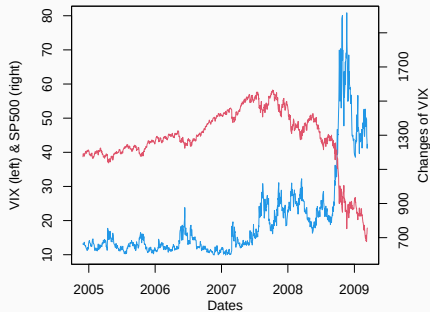
Stylized features of financial returns

Leverage effect

- Asset returns are negatively correlated with the changes of their volatilities (Black 1976, Christie 1982).
- As asset prices decline, companies become more leveraged (debt to equity ratios increase) and riskier, and hence their stock prices become more volatile.
- when stock prices become more volatile, investors demand high returns and hence stock prices go down.
- Volatilities caused by price decline are typically larger than the appreciations due to declined volatilities.

Stylized features of financial returns

VIX and SPY



Returns vs changes of volatility

